

Amanda Shum

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PROFESSIONAL SUMMARY

AI Engineer with experience in **Computer Vision, Deep Learning, Azure AI, workflow automation, and AI-assisted development, technical product management and project management. Microsoft Certified: Azure AI Apps and Agents Developer Associate.** Built convolutional neural network (CNN) and object detection systems using **PyTorch** and **TensorFlow**. Led cross-functional AI initiatives through project planning, stakeholder coordination, technical implementation, and **Agile delivery**. Combines hands-on engineering with technical leadership to help teams turn ideas into practical, responsible AI solutions.

EDUCATION

Simon Fraser University

Master of Professional Computer Science (Visual Computing)

Burnaby, BC

Expected Graduation Date: June 2027

Queen's University

Bachelor of Computing (Honours), Artificial Intelligence Specialization

Kingston, ON

June 2025

WORK EXPERIENCE

CanAI Garage, Shared Services Canada

Co-op Student (AI Engineering Lead)

Vancouver, BC

2026-Present

- Led a team of 4 to deliver a GitHub Copilot customization framework with 20+ prompts, instructions, skills, and engineering standards, aligning stakeholder requirements with business goals and Responsible AI practices. Standardized AI-assisted development and evaluation processes, reducing time spent writing and reviewing code by 50%.
- Configured GPT-4.1 model deployment, Azure Blob Storage, and Azure Function Apps, and owned technical planning for a public-sector retrieval-augmented generation (RAG) chatbot using Azure OpenAI and Azure AI Search, defining requirements for website ingestion, knowledge retrieval, metadata management and content refresh.
- Developed technical documentation, governance models, and training materials while coordinating sprint deliverables in Azure DevOps and translating stakeholder requirements into architecture, privacy, Responsible AI, and operational readiness considerations.

Kingston Student Housing Co-operative

Vice President

Kingston, ON

2021-2025

- Provided senior management and operational leadership across 21 residential properties, overseeing operations, human resources, facilities, member services, and safety compliance while advising the Board of Directors on budgets, policies, strategic planning, and revenue-generating initiatives.
- Led digital transformation by designing and deploying integrated booking, member application, document generation, and approval systems using Google Sheets, Google Apps Script, and HTML.
- Increased rental bookings by 25% through customer analysis, seasonal marketing, and room-allocation strategies; implemented Google Apps Script and Google Workspace automations that reduced administrative workload by 75% and supported responses to 10,000+ inquiries within 24 hours.

Queen's University

Teaching Assistant

Kingston, ON

2023-2025

- Supported undergraduate courses in Database Management Systems, Computer Architecture, and introductory Computer Science by delivering tutorials, grading assignments, managing course materials, facilitating forum discussions, and providing one-on-one guidance in SQL, relational database design, Python fundamentals, and problem-solving.

PROJECTS

Real-Time Pothole Detection and Relative Depth Estimation

Sept-Dec 2025

- Developed a **real-time pothole detection and relative-depth estimation pipeline** using *YOLOv8* and *MiDaS*, combining bounding-box region analysis, depth normalization, and noise filtering to distinguish potholes from surrounding road surfaces across varied lighting, textures, and camera perspectives.

KSHC Mailer and Workflow Automation System

2024-2025

- Designed and deployed an **automation system** for the Kingston Student Housing Cooperative using *Google Apps Script*, *Google Sheets*, and *HTML/JavaScript* for email communication, booking management, member applications, document generation, approval workflows, and role-based dashboards, improving data organization and reducing manual administrative processing.

Gridworld Coverage

Jan-Apr 2025

- Built and trained an RL agent using *Stable Baselines3*, *DQN*, *A2C*, and *PyTorch* in a custom Gridworld environment, optimizing reward functions and observation-space configurations to achieve an **85% area-coverage success rate**.

Fruit Ripeness Detection

Sept-Dec 2024

- Developed a two-stage deep learning system using *TensorFlow* and *NumPy*, combining a primary CNN for fruit classification with fruit-specific CNN models for ripeness prediction and achieving **96% fruit-classification accuracy**.

SKILLS

Programming Languages: Python, C, C++, JavaScript, PHP, SQL, R, HTML, CSS

Machine Learning and Computer Vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Ultralytics YOLOv8, Stable Baselines3, NumPy, pandas

AI and Cloud Technologies: Azure OpenAI, Azure Foundry, Azure Portal, Azure AI Vision, Azure AI Search, Azure Functions, Azure Blob Storage, Azure Content Understanding Studio, Azure Machine Learning Studio, Azure AI Document Intelligence, Google Workspace Studio, Retrieval-Augmented Generation

Development and Collaboration: Git, GitHub, Azure DevOps, Azure Boards, Visual Studio Code, Google Apps Script, SharePoint, Power BI, Power Automate, Power Apps

Core Competencies: Computer Vision, Image Processing, Deep Learning, Object Detection, Image Classification, Monocular Depth Estimation, Reinforcement Learning, Prompt Engineering, Context Engineering, Workflow Automation, Solution Architecture, Responsible AI, Agile Delivery, Product Planning, Project Coordination, Technical Documentation, Stakeholder Management

Additional Tools: Microsoft Office, Google Workspace, Microsoft Teams, Canva, Onshape, AnyDesk

Operating Systems: Windows, Linux

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)

Sept 2026